

Human-Computer Interaction — Lecture 01:

Introduction

المحاضرة 1: مقدمة في التفاعل بين الإنسان والحاسوب (HCI)

Multiple Choice (20)

1. What is Human-Computer Interaction (HCI)?

ما هو التفاعل بين الإنسان والحاسوب (HCI)؟

- A. The study of computer hardware manufacturing only
- B. A multidisciplinary field that focuses on the design, evaluation, and implementation of interactive computing systems for human use
- C. A programming language for building user interfaces
- D. A field that studies only network protocols

✓ **الإجابة الصحيحة:** B) A multidisciplinary field that focuses on the design, evaluation, and implementation of interactive computing systems for human use

ليه صح: HCI مجال متعدد التخصصات يركز على تصميم وتقييم وتنفيذ الأنظمة الحاسوبية التفاعلية لاستخدام الإنسان. الاختيارات الأخرى تحصره في مجال ضيق (عتاد أو لغة أو شبكات).

المرجع: Lecture 01 — Slide 3

2. Which question does the lecture raise about the abbreviation HCI?

إيه السؤال اللي بتطرحه المحاضرة عن اختصار HCI؟

- A. Whether the 'C' stands for Computer or Communication
- B. Whether the 'I' stands for Interaction or Interfaces
- C. Whether the 'H' stands for Human or Hardware
- D. Whether the abbreviation should be dropped entirely

✓ **الإجابة الصحيحة:** B) Whether the 'I' stands for Interaction or Interfaces

ليه صح: المحاضرة بتسأل: حرف I معناه Interaction ولا Interfaces؟ والدكتور كان بيظن إنها Interfaces. باقي الاختيارات مش مذكورة.

المرجع: Lecture 01 — Slide 4

3. According to the Oxford definition shown in the lecture, which devices mediate human-computer interaction?

حسب تعريف أكسفورد في المحاضرة، أي أجهزة يتوسط التفاعل بين الإنسان والحاسوب؟

- A. Only keyboards and mice
- B. Printers and scanners only
- C. Floppy disks and magnetic tapes
- D. Graphics devices, touch-sensitive devices, and voice-input devices

✓ **الإجابة الصحيحة:** D) Graphics devices, touch-sensitive devices, and voice-input devices

ليه صح: التعريف يذكر أجهزة الرسومات والأجهزة الحساسة للمس وأجهزة الإدخال الصوتي، ويؤكد على الإدخال/الإخراج مع البرمجيات المساندة.

المرجع: Lecture 01 — Slide 4

4. Which of the following is NOT listed as an example of HCI in daily life?

أي واحد من دول مش مذكور كمثال على HCI في الحياة اليومية؟

- A. Industrial robots welding car bodies
- B. ATM machines
- C. Voice assistants such as Siri and Alexa
- D. Car dashboards

✓ **الإجابة الصحيحة:** A) Industrial robots welding car bodies

ليه صح: الأمثلة المذكورة: الهواتف الذكية، ATM، لوحات السيارات، منصات التعلم، المساعدات الصوتية، المواقع والتطبيقات. روبوتات اللحام مش مذكورة.

المرجع: Lecture 01 — Slide 5

5. Which of the following is NOT stated as a result of good HCI?

أي واحدة مش مذكورة كنتيجة لـ HCI الجيد؟

- A. Better productivity
- B. Better user satisfaction
- C. Higher hardware manufacturing speed
- D. Lower training and support costs

✓ **الإجابة الصحيحة:** C) Higher hardware manufacturing speed

ليه صح: HCI الجيد يحسن الإنتاجية ورضا المستخدم ويقلل تكاليف التدريب والدعم. سرعة تصنيع العتاد مش منها.

المرجع: Lecture 01 — Slide 6

6. Why does the lecture say HCI matters in modern life?

ليه HCI مهم في الحياة الحديثة حسب المحاضرة؟

- A. It makes technology machine-centered to run faster
- B. It ensures technology is human-centered, not machine-centered
- C. It removes the need for interfaces
- D. It focuses on developers' convenience only

✓ الإجابة الصحيحة: B) It ensures technology is human-centered, not machine-centered

ليه صح: HCI بيضمن إن التكنولوجيا تكون human-centered مش machine-centered، وبيققل الأخطاء ويخلي الواجهات متاحة للجميع بمن فيهم ذوي الإعاقة.

المرجع: Lecture 01 — Slide 6

7. Which two fields feed into Human Factors Engineering in the hierarchy diagram?

أي مجالين بيغذوا Human Factors Engineering في المخطط؟

- A. Mathematics and Physics
- B. Engineering and Psychology
- C. Art and Marketing
- D. Networking and Databases

✓ الإجابة الصحيحة: B) Engineering and Psychology

ليه صح: المخطط بيوضح إن Engineering و Psychology هما المدخلات لـ Human Factors Engineering.

المرجع: Lecture 01 — Slide 7

8. Which fields appear directly under Human Factors Engineering in the diagram?

أي مجالات بتظهر مباشرة تحت Human Factors Engineering؟

- A. Industrial Design, HCI, and Product Design
- B. UI Design, UX Design, and Interaction Design
- C. Engineering, Psychology, and Sociology
- D. Databases, Networks, and Security

✓ الإجابة الصحيحة: A) Industrial Design, HCI, and Product Design

ليه صح: تحتته Industrial Design و HCI و Product Design. أما UI و UX و Interaction Design فهم تحت HCI.

المرجع: Lecture 01 — Slide 7

9. Which definition matches UX Design?

أي تعريف يطابق UX Design؟

- A. The process of designing buttons, colours, fonts, icons, and layouts
- B. The process of designing how the system responds to user actions only
- C. The process of designing products or systems to provide a good, easy, and satisfying experience for users
- D. The process of designing computer hardware circuits

✓ **الإجابة الصحيحة:** C) The process of designing products or systems to provide a good, easy, and satisfying experience for users

ليه صح: UX هو تصميم المنتجات لتوفير تجربة جيدة وسهلة ومُرضية. تصميم الأزرار والألوان والخطوط هو UI، وتصميم استجابة النظام لأفعال المستخدم أقرب لـ Interaction Design.

المرجع: Lecture 01 — Slide 8

10. Which definition matches UI Design?

أي تعريف يطابق UI Design؟

- A. The process of making the whole experience easy and satisfying
- B. The process of studying users' psychology in labs
- C. The process of choosing a database engine
- D. The process of designing the visual interface, including buttons, colours, fonts, icons, and layouts

✓ **الإجابة الصحيحة:** D) The process of designing the visual interface, including buttons, colours, fonts, icons, and layouts

ليه صح: UI هو تصميم الواجهة المرئية: الأزرار والألوان والخطوط والأيقونات والتخطيط. وصف جعل التجربة كلها سهلة ومُرضية هو تعريف UX.

المرجع: Lecture 01 — Slide 8

11. What does Interaction Design focus on?

إيه اللي بيركز عليه Interaction Design؟

- A. The colour palette of the interface
- B. The physical shape of the device
- C. How users interact with a system and how the system responds to their actions
- D. The cost of the product

✓ **الإجابة الصحيحة:** C) How users interact with a system and how the system responds to their actions

ليه صح: بيركز على طريقة تفاعل المستخدم مع النظام وكيف يستجيب النظام لأفعاله. الألوان من اختصاص UI.

المرجع: Lecture 01 — Slide 8

12. In a flight booking application, which concern belongs to UI Design?

في تطبيق حجز طيران، أي اهتمام يندرج تحت UI Design؟

- A. Making searching, selecting, and booking a flight easy and clear
- B. Whether the steps are clear and logical
- C. Whether users can achieve their goals easily
- D. The colour, size, shape, and position of the Book Now button

✓ الإجابة الصحيحة: D) The colour, size, shape, and position of the Book Now button

ليه صح: المثال في المحاضرة: UI يركز على لون وحجم وشكل ومكان زر Book Now. الباقي أسئلة UX.

المرجع: Lecture 01 — Slide 9

13. Which question is typical of UX Design?

أي سؤال نموذجي لـ UX Design؟

- A. Which font size should the menu use?
- B. What colour should the icons be?
- C. How should the layout be arranged on the page?
- D. Can users find what they need easily?

✓ الإجابة الصحيحة: D) Can users find what they need easily?

ليه صح: UX ببساطة: هل التطبيق سهل؟ هل المستخدم يحقق أهدافه؟ هل يلاقي اللي محتاجه؟ أما الخط والألوان والتخطيط فهي UI.

المرجع: Lecture 01 — Slide 9

14. Which of the following is listed among the HCI design principles in the lecture?

أي واحد من دول مذكور ضمن مبادئ تصميم HCI في المحاضرة؟

- A. Redundancy
- B. Tolerance
- C. Latency
- D. Encryption

✓ الإجابة الصحيحة: B) Tolerance

ليه صح: الشريحة بتعرض Consistency و Mapping و Constraints و Feedback و Discoverability و Affordances و Structure و Simplicity و Tolerance و Equity و Flexibility و Perceptibility و Ease و Documentation و Comfort.

المرجع: Lecture 01 — Slide 10

15. How many HCI design principles appear in the lecture's figure?

كام مبدأ تصميم HCI ظاهر في شكل المحاضرة؟

- A. 10
- B. 12
- C. 20
- D. 15

✓ الإجابة الصحيحة: D) 15

ليه صح: الشكل فيه 3 صفوف $\times$ 5 مبادئ = 15 مبدأ.

المرجع: Lecture 01 — Slide 10

16. Which list contains the five main components of HCI?

أي قائمة فيها المكونات الخمسة الرئيسية لـ HCI؟

- A. Users, Tasks, Tools, Interfaces, Context
- B. Users, Data, Servers, Networks, Security
- C. Hardware, Software, Peopleware, Data, Network
- D. Input, Process, Output, Storage, Feedback

✓ الإجابة الصحيحة: A) Users, Tasks, Tools, Interfaces, Context

ليه صح: المكونات: Users و Tasks و Tools و Interfaces و Context. الباقي قوائم لمفاهيم مختلفة.

المرجع: Lecture 01 — Slide 11

17. In the HCI components model, what is the role of Interfaces?

إيه دور Interfaces في مكونات HCI؟

- A. The connection between users and tools
- B. The environment where interaction occurs
- C. The activities users aim to complete
- D. The people interacting with technology

✓ الإجابة الصحيحة: A) The connection between users and tools

ليه صح: Interfaces = الربط بين المستخدمين والأدوات. البيئة هي Context، والأنشطة هي Tasks، والأشخاص هم Users.

المرجع: Lecture 01 — Slide 11, 14

18. Why do designers create various tasks and scenarios?

ليه المصممين بينشئوا مهام وسيناريوهات متنوعة؟

- A. To understand the user journey
- B. To reduce the cost of servers
- C. To measure CPU temperature
- D. To replace human users

✓ الإجابة الصحيحة: A) To understand the user journey

ليه صح: عشان يفهموا رحلة المستخدم (user journey). باقي الاختيارات ملهاش علاقة.

المرجع: Lecture 01 — Slide 13

19. Why do researchers spend time understanding the needs and expectations of the target audience?

ليه الباحثين بيقضوا وقت في فهم احتياجات الجمهور المستهدف؟

- A. To reduce hardware costs
- B. To increase processor speed
- C. To create a user-centered design that ensures usefulness and usability
- D. To avoid testing the product

✓ الإجابة الصحيحة: C) To create a user-centered design that ensures usefulness and usability

ليه صح: فهم الاحتياجات ببساعد في تصميم user-centered يضمن الفائدة وسهولة الاستخدام.

المرجع: Lecture 01 — Slide 12

20. According to the lecture, the tools used to create technology need to be robust and flexible enough to allow for what?

حسب المحاضرة، الأدوات لازم تكون قوية ومرنة كفاية عشان تسمح بإيه؟

- A. Hardware replacement
- B. Network deployment
- C. Iterations
- D. Legal compliance

✓ الإجابة الصحيحة: C) Iterations

ليه صح: لازم تسمح بالتكرار (iterations) في التصميم.

المرجع: Lecture 01 — Slide 13

Essay and Calculation Questions (8)

1. Define HCI and explain what it studies.

عرّف HCI وشرح إيه اللي بيدرسه.

Model answer:

HCI is a multidisciplinary field that focuses on the design, evaluation, and implementation of interactive computing systems for human use.

It also studies the major phenomena surrounding these systems:

- how people interact with computers,
- how systems respond,
- and how interfaces can be improved for better usability.

المرجع: Lecture 01 — Slide 3

2. Explain the discussion about whether the 'I' in HCI means Interaction or Interfaces, and summarize the Oxford definition shown.

اشرح النقاش حول حرف I في HCI ولخص تعريف أكسفورد.

Model answer:

The lecture asks whether the I is for Interaction or Interfaces; the lecturer used to think it was Interfaces, and the abbreviation possibly adapted to the better word.

The Oxford definition describes the human-computer interface as the means of communication between a human user and a computer system, referring to input/output devices with supporting software.

Devices mentioned: graphics devices, touch-sensitive devices, and voice-input devices, configured to facilitate an efficient and desirable interaction.

المرجع: Lecture 01 — Slide 4

3. Give examples of HCI in daily life and explain why HCI matters.

اذكر أمثلة على HCI في الحياة اليومية وشرح ليه HCI مهم.

Model answer:

Examples: smartphone interfaces, ATM machines, car dashboards, e-learning platforms, voice assistants (Siri, Alexa), websites and mobile applications. Every button, menu, icon, and gesture is influenced by HCI principles.

Why it matters: users can interact with technology easily; systems reduce errors and confusion; interfaces are accessible to all, including people with disabilities; technology becomes human-centered, not machine-centered.

Good HCI leads to better productivity, better user satisfaction, and lower training and support costs.

المرجع: Lecture 01 — Slide 5, 6

4. Describe the relation between HCI and other fields using the hierarchy in the lecture.

اوصف علاقة HCI بالمجالات الأخرى حسب المخطط الهرمي في المحاضرة.

Model answer:

Engineering and Psychology feed into Human Factors Engineering.

Human Factors Engineering leads to Industrial Design, HCI, and Product Design.

HCI in turn includes UI Design, UX Design, and Interaction Design.

المرجع: Lecture 01 — Slide 7

5. Differentiate between UX Design, UI Design, and Interaction Design. Use the flight booking example.

فرّق بين UX وUI وInteraction Design مع مثال حجز الطيران.

Model answer:

UX Design: designing products or systems to give a good, easy, and satisfying experience; it asks whether the app is easy to use, whether users achieve goals, whether steps are clear and logical, and whether users find what they need and feel comfortable.

UI Design: designing the visual interface: menus, font size and colour, buttons, layout, icons.

Interaction Design: designing how users interact with the system and how the system responds to their actions.

Flight booking example: UX makes searching, selecting, and booking easy and clear; UI focuses on the colour, size, shape, and position of the Book Now button.

المرجع: Lecture 01 — Slide 8, 9

6. List the principles of human-computer interaction design shown in the lecture.

اذكر مبادئ تصميم HCI المعروضة في المحاضرة.

Model answer:

Discoverability, Feedback, Constraints, Mapping, Consistency,

Affordances, Structure, Simplicity, Tolerance, Equity,

Flexibility, Perceptibility, Ease, Comfort, Documentation.

المرجع: Lecture 01 — Slide 10

7. Explain the five main components of HCI.

اشرح المكونات الخمسة الرئيسية لـ HCI.

Model answer:

Users: people interacting with technology; researchers study their needs and expectations to create user-centered design ensuring usefulness and usability.

Tasks: activities users aim to complete; designers create tasks and scenarios to understand the user journey.

Tools: devices and software enabling tasks; they must be robust and flexible enough to allow iterations.

Interfaces: the connection between users and tools, helping users navigate the design effectively to accomplish goals.

Context: the environment where interaction occurs and how it affects the way the user interacts with the system.

المرجع: Lecture 01 — Slide 11-14

8. Summarize the conclusion of Lecture 1.

لخص خلاصة المحاضرة الأولى.

Model answer:

HCI focuses on the interaction between people and computers.

It aims to make systems easy, useful, efficient, and satisfying to use.

It is important because good design improves user experience and reduces errors.

It is applied in mobile applications, websites, healthcare systems, education, and automotive systems.

المرجع: Lecture 01 — Slide 15